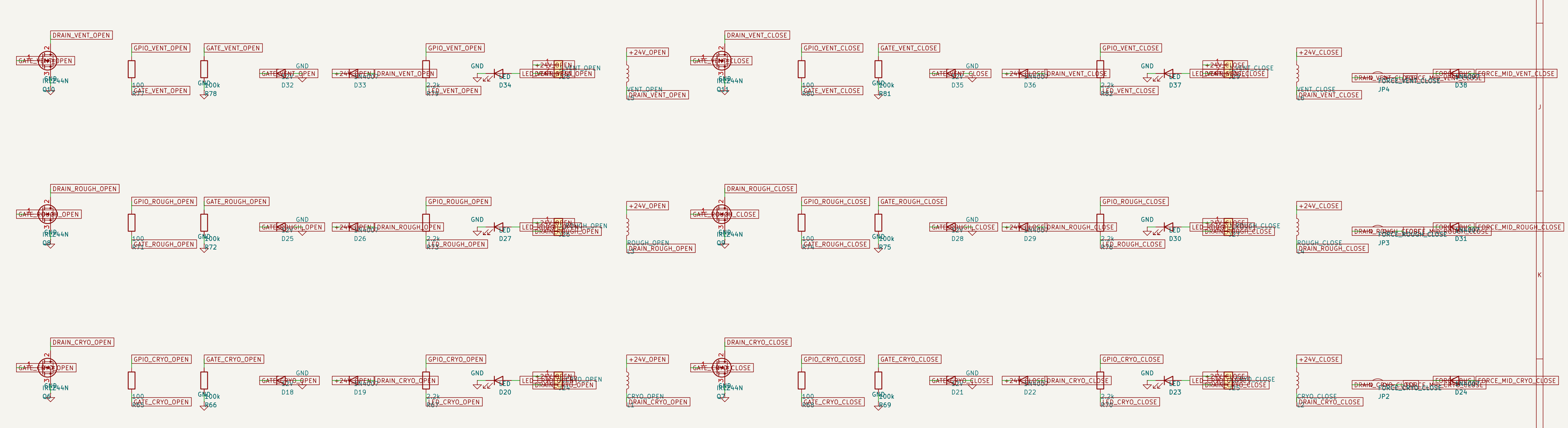
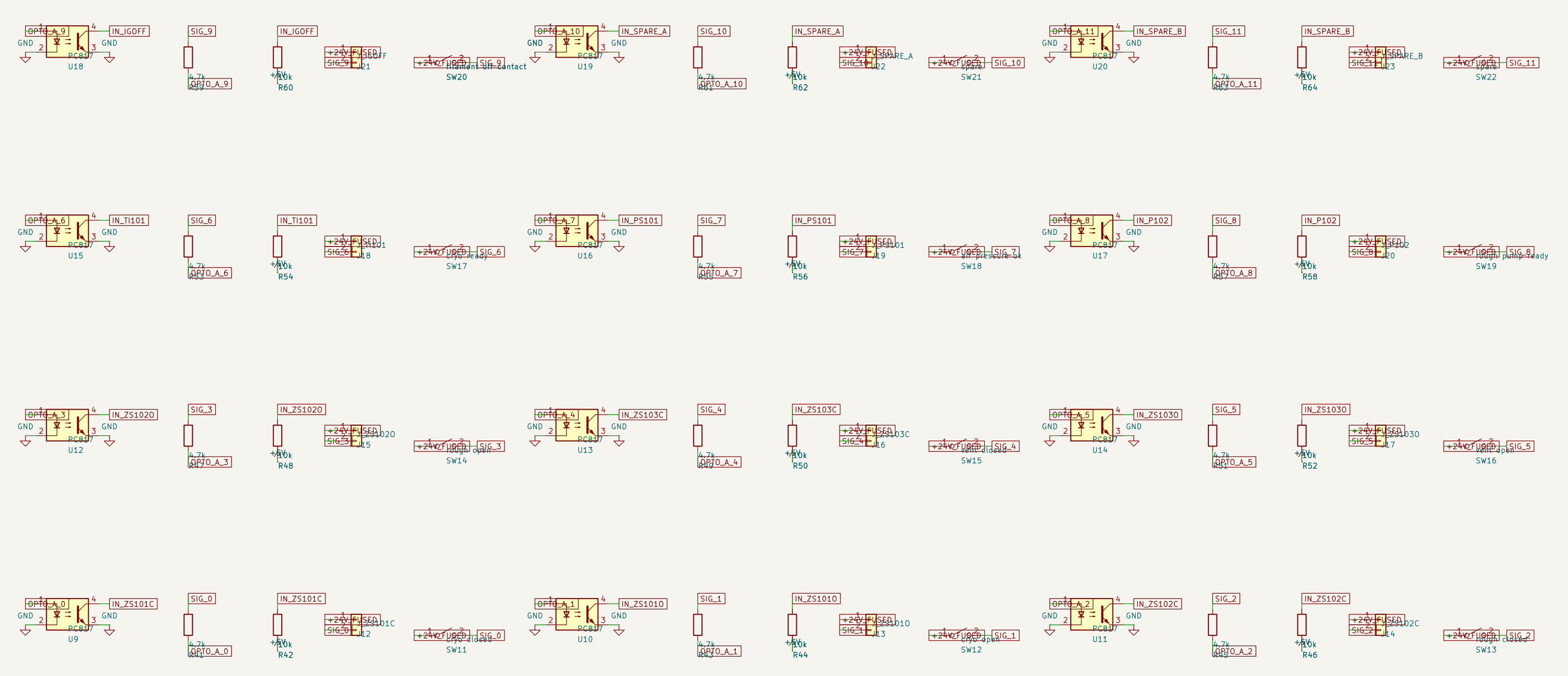
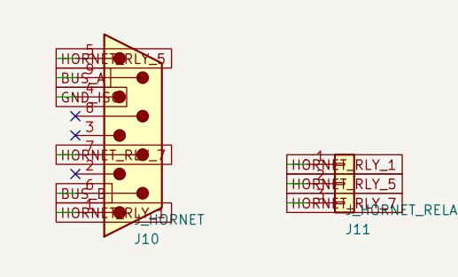
OFF BOARD ON PURPOSE
Cryo–Torr 8, CTI 8200 helium compressor, and IDP–15 are not on this copper.
This board asks valves to move and reads proofs. It does not start a pump motor or a compressor.
Before any coil is landed: meter K2 unpowered. Pin 2 must conduct to pin 4 and not to pin 3.



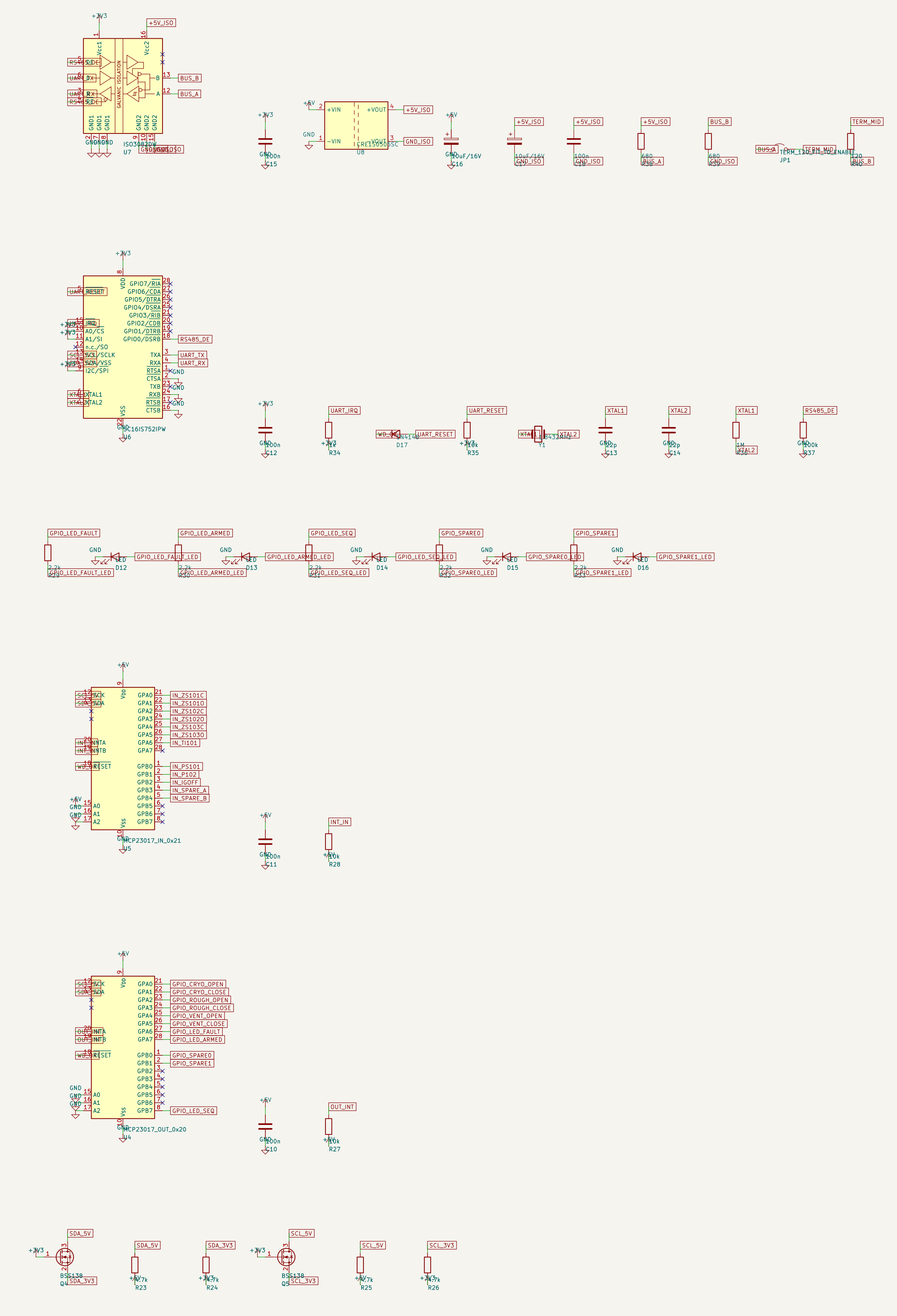
COIL DRIVERS low–side IRLZ44N, flyback on the terminal, gate clamped.
OPEN coils take the inhibited 24 V rail. CLOSE coils take the fused rail that stays up.
A close channel's force diode is armed only when its jumper shunt is fitted.



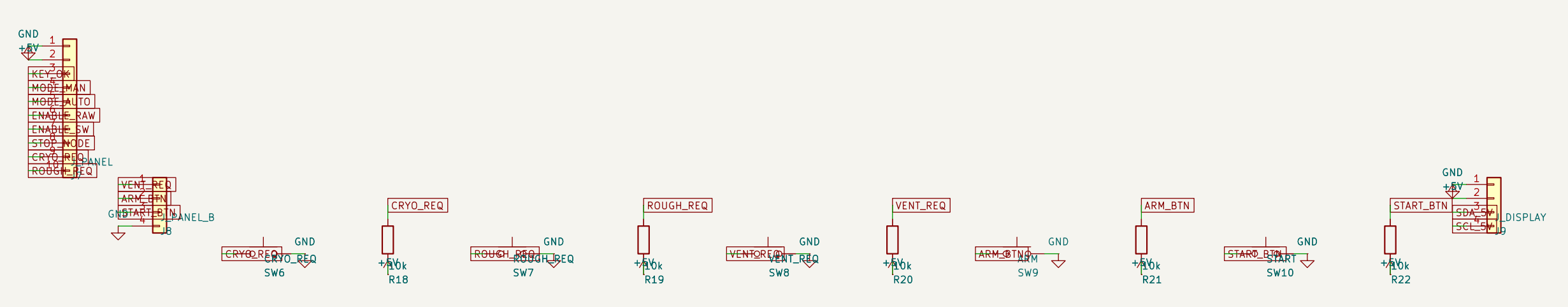
CONTACT INPUTS +24 V through 4.7k and the opto LED. Closed contact = expander pin low = true.
Spares are landings only. They are not a turbo interlock.



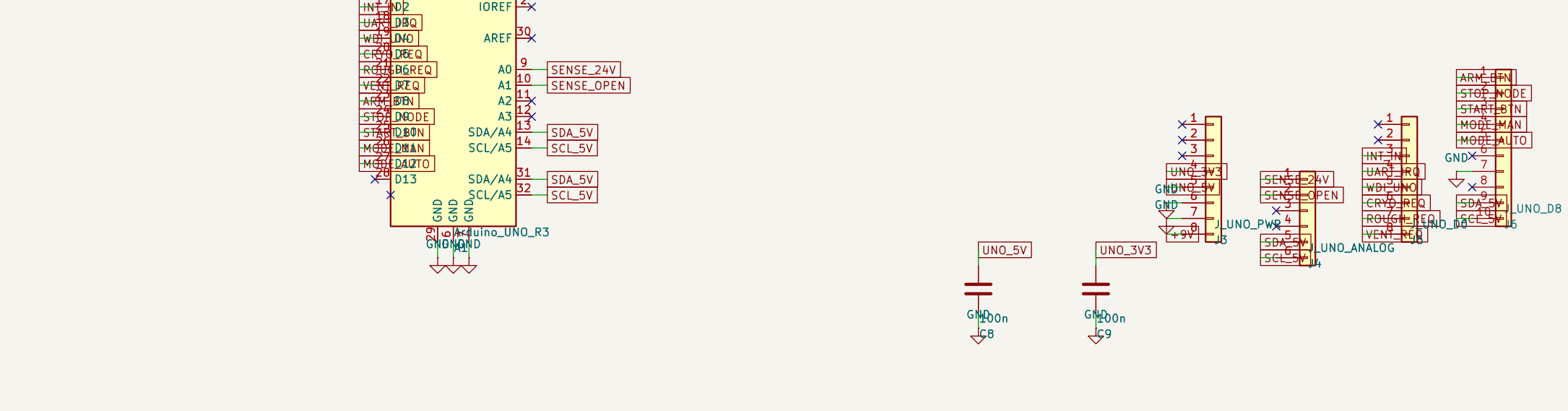
HORNET DE–9 MALE plugs into the gauge's female RS–485 socket.
A of the ISO3082 (non–inverting) lands on pin 9. B lands on pin 6. Swap only if the idle bus is backwards.
Pin 4 is isolated ground. Do not tie it to Uno ground. Pin 2 is not connected.



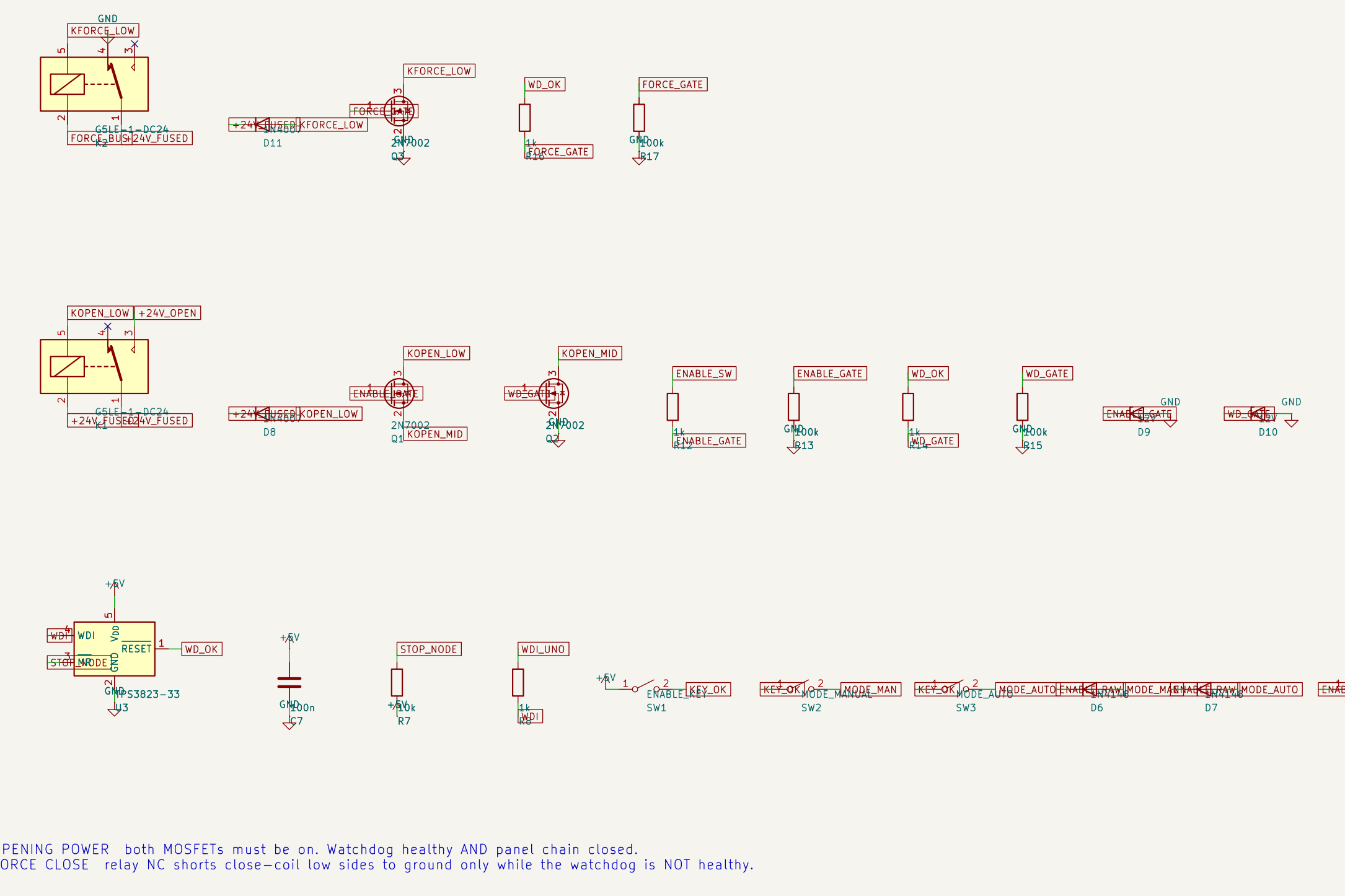
I2C 5 V expanders, 3.3 V UART. Pull–ups on both sides of the shifter.



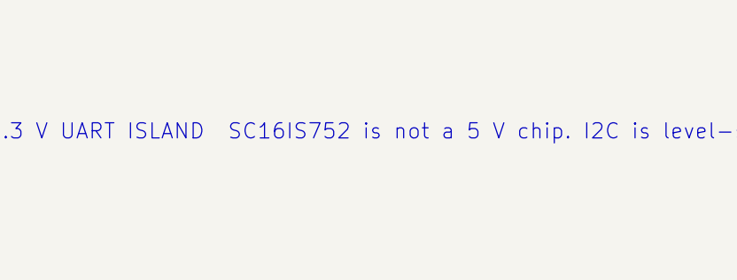
PANEL momentary requests to ground. Mode and enable pass 5 V only as far as the relay gates.



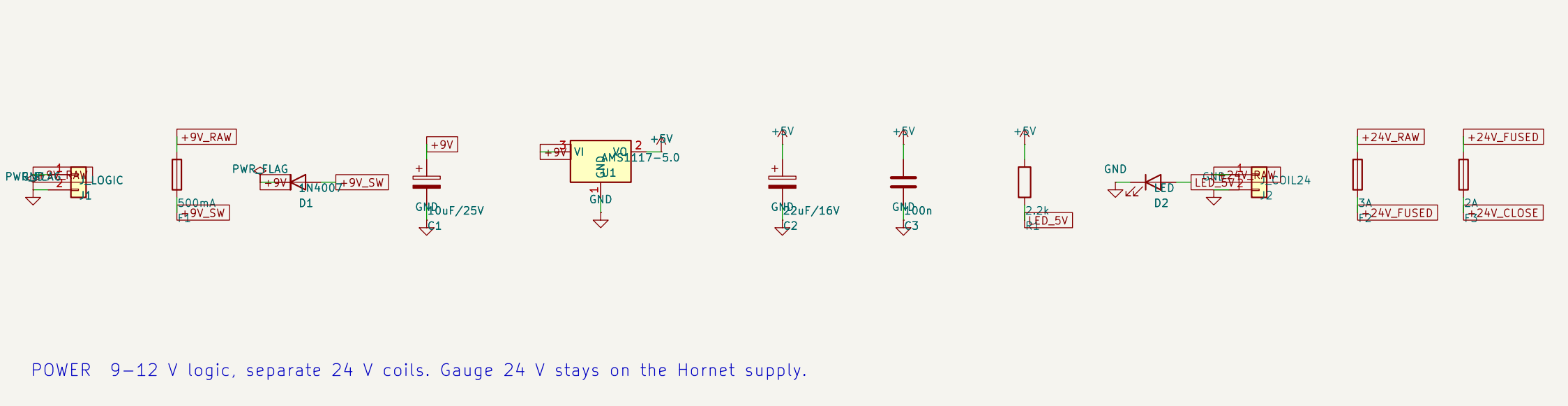
UNO R3 sockets are the fabricated part. The Uno symbol is the plugged–in board, not a copper part.



OPENING POWER both MOSFETs must be on. Watchdog healthy AND panel chain closed.
FORCE CLOSE relay NC shorts close–coil low sides to ground only while the watchdog is NOT healthy.



3.3 V UART ISLAND SC16IS752 is not a 5 V chip. I2C is level–shifted.



POWER 9–12 V logic, separate 24 V coils. Gauge 24 V stays on the Hornet supply.

MAIZE PHASE 5 CONTROLLER – read this before believing a wire
Assumptions, because the gate, pilot, and sensor terminals are still unmeasured:
1. Coil drivers are 24 V DC, about 0.5 A each. An AC coil does not belong on this board.
2. Opening power is a relay contact. It drops if the processor stops kicking the watchdog,
   the enable key is open, the mode is OFF, or STOP is pressed.
3. Close–coil force jumpers stay OPEN until a double–acting actuator is proven.
   Spring–return valves must leave those jumpers empty. Loss of 24 V then lets the spring close.
4. Sensor inputs are dry contacts. The board puts about 5 mA through a closed contact.
5. Hornet DE–9 follows the archived IGM402 map: 9 = data B (+), 6 = data A (–), 4 = isolated reference.
   Pin 2 (gauge 20–28 V) is not connected. Relay pins 1, 5, 7 are brought out and are not a filament proof.
6. Cold–ready, filament–off, and pump–ready are contact landings waiting for the real measurement.
7. The scroll pump, Cryo–Torr, and CTI compressor are not powered or started by this board.
8. Logic input is 9–12 V at J_LOGIC. Do not also power the Uno from USB.
9. UART 7–bit I2C address is 0x48 (A1 and A0 tied to 3.3 V). Baud is a firmware choice after the Hornet manual matches

Do not apply USB power and the 9 V logic input together.
RS–485 ground is isolated from Uno ground.
24 V DC coils only. Hornet pin 2 is not connected.
Design assumptions. Not a measured field pinout.
PPML
Sheet: /
File: maize–controller.kicad_sch
Title: MAIZE phase 5 controller
Size: User
Date: 2026–10–20
Rev: A
KiCad E.D.A. 10.0.7
Id: 1/1